

# IPL Dataset 2008-25

## Analysis Report

Exploratory Data Analysis using Python & Pandas

### 1. Problem Statement

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#### Background

You are working as a Data Analyst for a sports analytics firm. The firm has collected ball-by-ball delivery data from the Indian Premier League (IPL) spanning from 2008 to 2025. The goal is to extract meaningful insights about team performance, player statistics, and match dynamics to support analysts, commentators, and franchise strategists.

You are provided with a dataset (deliveries.csv) containing ball-by-ball records across all IPL seasons.

The dataset includes batting team, bowling team, batsman, bowler, runs, extras, wickets, and over details.

#### IPL Dataset in Questions to Answer

Your analysis should answer the following IPL Dataset in questions clearly.

1. Which team batting first gave the highest target?
2. Which team took the most wickets in the powerplay?
3. Which team lost more wickets in the power play?
4. Which team has set the 200+ target more times?
5. How many times has Z Khan dismissed SC Ganguly?
6. Who is the highest wicket taker for RCB?

7. Which player has taken the most wickets in IPL so far?
8. Which team has given the highest extra runs in IPL so far?
9. Which bowler has taken the highest wickets for each team? (Chart)

## 2. Solution Overview

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We will solve this problem using Exploratory Data Analysis (EDA).

EDA helps us understand patterns, relationships, and anomalies in the data before applying any advanced modeling.

The solution is divided into logical steps:

1. Data loading and understanding
2. Data cleaning and preparation
3. Analysis to answer each business question
4. Summary of insights

Each step below explains what we are doing and why.

## 3. Data Loading and Initial Understanding

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### Step 1: Import Required Libraries

We use Pandas for data manipulation and Matplotlib for visualization.

```
from itertools import groupby
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

### Step 2: Load the Dataset

**Why this step matters:**

- Allows us to inspect the structure of the dataset
- Helps verify column names and data types

```
df = pd.read_csv("deliveries.csv")
df.head()
```

**Step 3: Basic Dataset Overview****Why this step matters:**

- Identifies number of rows and columns
- Reveals Dataset Basic Information like Number of Rows And Columns, Columns Name With Datatype And NULL Values, Memory usage
- Reveals missing values and incorrect data types

```
df.shape
df.info()
print(df.isna().sum())
```

## 4. Data Cleaning and Preparation

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Real-world datasets are messy. Cleaning ensures accurate analysis.

**Step 4: Standardize Column Names****Why:**

- Makes column names consistent and easier to use in code

```
df.columns =
df.columns.str.strip().str.lower().str.replace(' ', '_')
print(df.columns.tolist())
```

## Step 5: Remove Duplicates

### Why:

- Duplicate rows can skew averages and insights

```
df.drop_duplicates(inplace=True)
```

## 5. Answering IPL Dataset with Analysis

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### Question 1: Which team batting first gave the highest target?

#### Why:

- Identifies the highest scoring first innings across all matches

```
targets = df[df['inning'] == 1].groupby(['match_id',
    'batting_team'])['total_runs'].sum()
max_target_value = targets.max()
max_target_team = targets.idxmax()[1]
max_match_id = targets.idxmax()[0]
print(f' The team batting first in match_id {max_match_id}
    target of {max_target_value} is: {max_target_team}')
```

### Question 2: Which team took the most wickets in the powerplay?

#### Why:

- Powerplay (overs 1-6) wickets indicate early dominance and bowling strength

```
powerplay_wickets = df[df['over'] <=
    5].groupby('bowling_team')['is_wicket'].sum()
max_wickets_team = powerplay_wickets.idxmax()
print(f'The team that took the most wickets in the
    powerplay is: {max_wickets_team}')
```

**Question 3: Which team lost more wickets in the power play?****Why:**

- Reveals batting fragility in the crucial opening phase

```
powerplay_wickets_lost = df[df['over'] < 6].groupby('batting_team')['is_wicket'].sum()
max_wickets_lost_team = powerplay_wickets_lost.idxmax()
print(f'The team that lost more wickets in the power play is: {max_wickets_lost_team}')
```

**Question 4: Which team has set the 200+ target more times?****Why:**

- 200+ targets reflect consistently aggressive batting lineups

```
targets = df[df['inning'] == 1].groupby(['match_id', 'batting_team'])['total_runs'].sum()
targets_200_plus = targets[targets >= 200]
target_counts = targets_200_plus.groupby(level=1).size()
max_target_team = target_counts.idxmax()
print(f'Team set 200+ target most: {max_target_team} with {target_counts.max()} times.')
```

**Question 5: How many times has Z Khan dismissed SC Ganguly?****Why:**

- Head-to-head matchup data helps understand player rivalries

```
dismissals = df[(df['bowler'] == 'Z Khan') & (df['player_dismissed'] == 'SC Ganguly')]
num_dismissals = dismissals.shape[0]
print(f'Z Khan has dismissed SC Ganguly {num_dismissals} times.')
```

**Question 6: Who is the highest wicket taker for RCB?****Why:**

- Team-specific bowling stats highlight franchise bowling pillars

```
rcb_wickets = df[df['bowling_team'] == 'Royal Challengers Bangalore']  
.groupby('bowler')['is_wicket'].sum()  
max_wickets_bowler = rcb_wickets.idxmax()  
print(f'The highest wicket taker for RCB is:  
{max_wickets_bowler}')
```

```
print(f'with {rcb_wickets.max()} wickets.')
```

**Question 7: Which player has taken the most wickets in IPL so far?****Why:**

- All-time wicket leader highlights the greatest IPL bowler

```
wicket = df.groupby('bowler')['is_wicket'].sum()  
ipl_highest_wicket_bowler = wicket.idxmax()  
print(f'Highest IPL wicket taker:  
{ipl_highest_wicket_bowler}')
```

```
print(f'Total wickets: {wicket.max()}')
```

**Question 8: Which team has given the highest extra runs in IPL?****Why:**

- Extra runs reveal bowling discipline issues across franchises

```
Extra =  
df.groupby('bowling_team')['extra_runs'].sum()  
IPL_Highest_Extra_run_Team = Extra.idxmax()  
print(f'Most extras conceded:  
{IPL_Highest_Extra_run_Team}')
```

```
print(f'Extra Runs: {Extra.max()}')
```

## 6. Deeper Insights — Chart Analysis

**Question 9: Which Bowler has taken the highest wickets for each team?**

**Why:**

- Visual comparison reveals top performers per franchise
- Bar chart with annotated bowler names provides quick reference

```
wickets_by_bowler_team = df.groupby(['bowling_team',
    'bowler'])['is_wicket'].sum().reset_index()
top_bowlers=wickets_by_bowler_team.sort_values('is_wicket',
    ascending=False) .groupby('bowling_team').head(1)

plt.figure(figsize=(10, 5))
plt.bar(top_bowlers['bowling_team'],
    top_bowlers['is_wicket'], color='skyblue')

for index, row in top_bowlers.iterrows():
    plt.text(row['bowling_team'], row['is_wicket'],
    row['bowler'],
        ha='center', va='bottom', fontsize=5,
    color='blue')

plt.xlabel('Bowling Team')
plt.ylabel('Wickets Taken')
plt.title('Highest Wicket Taker for Each Team in IPL')
plt.xticks(rotation=75, fontsize=8, color='Green')
plt.tight_layout()
plt.show()
```

## Dataset Column Reference

| Column Name      | Data Type | Description                        |
|------------------|-----------|------------------------------------|
| match_id         | Integer   | Unique identifier for each match   |
| inning           | Integer   | Inning number (1 or 2)             |
| batting_team     | String    | Name of the batting team           |
| bowling_team     | String    | Name of the bowling team           |
| over             | Integer   | Over number (0-based)              |
| ball             | Integer   | Ball number within the over        |
| batsman          | String    | Batsman on strike                  |
| bowler           | String    | Bowler delivering the ball         |
| total_runs       | Integer   | Total runs scored off the delivery |
| extra_runs       | Integer   | Extra runs (wides, no-balls, etc.) |
| is_wicket        | Integer   | 1 if wicket fell, 0 otherwise      |
| player_dismissed | String    | Name of dismissed player           |

## 7. Final Insights Summary

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From the analysis of the IPL 2008-25 deliveries dataset, we can conclude:

- Powerplay phases are crucial — teams that take early wickets consistently dominate matches
- A few star bowlers account for a disproportionate share of each franchise's wickets
- Extra runs are a significant factor — teams with bowling discipline concede fewer extras
- 200+ targets are rare achievements that distinguish the most explosive batting lineups
- Historical head-to-head records reveal clear player-level rivalries and dominances
- The highest wicket taker reflects sustained excellence across multiple seasons



**These insights can guide franchise management, auction strategies, and match-day tactics.**